

KOUSHIK GUDIPELly

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EDUCATION

- **NC State University** - Master of Computer Science [**GPA: 3.8/4.00**] **Aug 2023 - May 2025**
Coursework: Data structures & Algorithms, Software Engineering, Database Management Systems, Object Oriented design & Development, Neural Networks, Computer Networks, Automated Learning and Data Analysis, Privacy in the Digital Age, Social Computing and DAI.
- **VNRVJIET** - Bachelor of Technology in Computer Science [GPA: 8.64/10.00] **Jun 2018 - May 2022**

TECHNICAL SKILLS

- **Programming/ Scripting Languages:** Python, C++, Java, Javascript, Ruby, C, Typescript.
- **Web Technologies:** NodeJS, ReactJS, HTML, CSS, Bootstrap, GIT, Ruby on Rails.
- **Databases and Tools:** MySQL, MongoDB, Docker.
- **DataScience and Cloud Platforms:** Machine Learning, Deep Learning and Neural Networks, AWS Cloud.

WORK EXPERIENCE

Analyst - Full Time, Deloitte USI, Hyderabad, India **Jun 2022 - Jul 2023**

- Spearheaded the development of the DAS AM project, focusing on elevating Deloitte's data management and analysis capabilities across multiple projects through strategic implementation of Angular and .NET technologies.
- Engineered and maintained sophisticated web interfaces, ensuring seamless integration with .NET backend systems, which significantly enhanced data handling processes including input, retrieval, and analysis.
- Delivered intuitive user interface designs that substantially improved the user experience, fostering more efficient data interaction and visualization.
- Played a crucial role in cross-functional team collaborations, adopting Agile methodologies to facilitate timely and successful project milestones achievements, thereby boosting the effectiveness of Deloitte's data-driven strategies.

PROJECTS

Web Application Development (Police Database For Women Safety)

- Developed a comprehensive and reliable database using MySQL DB, Java, HTML, CSS, Bootstrap, JSP, and Servlets to store and manage data related to women's safety in the local area. This system provided essential data to the local police department, actively monitored to enhance women's safety.
- Created an application that tracks women's locations in real-time and checks for any unusual activity patterns. If an unusual location is detected, the system sends a notification to the local police department after prior verification, ensuring timely intervention and support.

Machine Learning (Privacy Preserving AI/ML In Chat Apps)

- Led the development of a privacy-preserving AI/ML framework using Python, ML, PySyft, PyTorch, and AWS EC2 instances. The project's main objective was to enable Federated Learning to ensure data privacy while allowing machine learning on private data.
- Implemented remote machine learning capabilities to analyze and mitigate cyberbullying in chat applications without compromising user privacy. This solution allows businesses and researchers to harness valuable data insights while adhering to strict privacy standards.

Web Application Development (Framework for Private ML)

- Designed and developed a sophisticated framework using Spring Boot, PostgreSQL, Python, FastAPI, and ReactJS to enable AI/ML on private data through federated learning. This approach ensures that sensitive data remains confidential while still allowing machine learning processes.
- Constructed a multi-tier architecture to facilitate seamless expansion and deployment of the framework. The architecture includes a Python library for core functionalities, a Spring Boot server for backend operations, a FastAPI-based microservice for handling machine learning tasks, and a ReactJS web interface for project registration and management.
- Led a team of four in executing the project, providing guidance and oversight to ensure proper implementation of the framework. The project allows developers to easily incorporate privacy-preserving ML into their applications, addressing the need for secure and efficient data processing.

Machine Learning / Deep Learning (Emotion Detection System)

- Devised a novel emotion detection system employing deep learning techniques to accurately recognize seven categories of human emotions through facial expressions, capable of real-time predictions.
- Trained the application on an extensive dataset, using a highly optimized Convolutional Neural Network and VGG 16 models, achieving remarkable training and validation accuracies of 97.3% and 95.7%, respectively.